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Asymmetric Nash bargaining model for operation optimization of multi-integrated energy systems considering peer-to-peer energy trading

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ABSTRACT

Energy interactions across integrated energy systems constitute crucial means to enhance energy efficiency and match supply and demand. However, the cooperative operation and benefit distribution among multiple integrated energy systems still need in-depth exploration. To fill this gap, this paper presents an asymmetric Nash bargaining optimization model for multiple integrated energy systems considering peer-to-peer trading. First, a scheduling model for multi-integrated energy systems is formulated considering carbon trading. Then, the model is incorporated into Nash bargaining framework and transformed into the alliance cost minimization subproblem and peer-to-peer trading payment bargaining subproblem. The bargaining power factor is introduced to measure the contribution of participants in energy sharing. The alternating direction multiplier method is utilized to handle the proposed model. Finally, a case study is carried out to validate the validity of the proposed strategy. The results show that compared with the independent operation mode, the performances of three integrated energy systems in collaborative operation mode are enhanced by 11.7 %, 9.0 %, and 4.8 % respectively. The distributed algorithm can reduce the computation time by 30 % and obtain highly efficient solutions while protecting private information of each participant. This research provides support and practical tools for conducting peer-to-peer transactions of multiple integrated energy systems.

1. Introduction

In the past decades, global energy consumption has witnessed unprecedented growth in line with economic and technological evolution (Karaca et al., 2023). The massive utilization of fossil fuels has exacerbated environmental pollution and created an energy crisis. To realize the sustainability of society, economy and environment, the 17 sustainable development goals (SDGs 17) were adopted by the United Nations in 2015. To attain SDG 17 by 2030, investments in photovoltaic and wind energy are imperative, alongside improving energy productivity. In this context, a modern energy system emphasizing cleanliness, lower carbon emissions, safety, and efficiency has become central to sustainable development (Yi & Ren, 2024). The integrated energy system (IES) leverages renewable energies comprehensively, facilitating seamless coordination among various forms of energy. Specifically, the IES internally couples various forms of energy and, by flexibly utilizing energy production, energy conversion, and storage equipment, achieves highly efficient complementarity among multiple energies (Chen et al., 2023; Yang & Liu, 2024). Advances in energy network technologies have enabled networks of multiple IESs to engage in peer-to-peer (P2P) energy trading, reducing reliance on centralized energy networks (Wang et al., 2024b). Multi-IESs can realize energy sharing, promote comprehensive utilization of renewable energy, and rationally allocate resources among IESs to achieve better economic benefits. However, each IES is governed by different stakeholders, and it is of great significance to develop a benefit distribution framework. Hence, optimizing the operational models and equitably distributing benefits across multi-IESs are pivotal research priorities. This research aims to introduce energy sharing among IESs to realize P2P trading, investigate the shared energy trading mechanism and further propose a benefit distribution framework to incentivize active participation of IESs in P2P energy transactions.

For the optimized operation of an individual IES, many achievements have already been reached currently (Zhang et al., 2023). The operational challenges faced by individual IES usually cover demand response, uncertainty, and multi-stakeholder operation. Regarding demand response and equipment changeable operation conditions, Wang et al. (2024a) formulated an innovative multi-timescale optimal scheduling strategy for the IES. The research indicates their presented

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Nomenclature

Abbreviation

AC	absorption chiller
ADMM	alternating direction method of multiplier
CHP	combined heat and power
EB	electric boiler
EC	ELECTRIC chiller
ES	energy storage system
EES	electrical storage system
GB	gas boiler
HES	heating storage system
IES	integrated energy system
PV	photovoltaic
P2P	peer-to-peer
SNB	standard Nash bargaining
SNB	standard Nash bargaining
WHB	waste heating boiler
WT	wind turbine

Subscripts

i/j	the ith/jth IES
k	number of iterations
min/max	minimum/maximum
net	network-crossing charges
η_{WHB} /η_{GB}	efficiency of WHB/GB
η_{e-h}	efficiency of electricity-heat
$\eta_{ES}^{ch}/\eta_{ES}^{dc}$	charging/discharging efficiency of ES
$\lambda_{i-j,t}^{e,P2P}/\lambda_{i-j,t}^{h,P2P}$	P2P trading price of electricity/heating between IES i and IES j (CNY/kWh)
$\lambda_{j-i,t}^{e,*}/\lambda_{j-i,t}^{h,*}$	desired trading electricity/ heating price for IES j (CNY/kW
π	carbon price (CNY/t)
$\rho_i^{e,SP}/\rho_i^{h,SP}$	penalty factor of subproblem
ϕ_i	bargaining power factor
ξ	dual residual
σ	original residual
$\tau_{grid}/\tau_{GB}^{h}$	actual carbon emission factor of grid/GB (kg/kWh)
$\tau_{CHP}^{e}/\tau_{CHP}^{h}$	actual carbon emission factor of CHP (kg/kWh)

Latin symbols

A_{PV}	installation scale of PV (m^2)
$c_{i,s}$	unit maintenance costs of equipment s in IES i (CNY/kW)
C_i^{IO}	independent operation cost of IES i (CNY)
C_i^{CO}	cooperative operation cost of IES i (CNY)
$C_{i,buy}$	energy purchased cost of IES i (CNY)
$P_{AC,t}^{C}/P_{AC,t}^{h}$	cooling/heating power of AC(kW)
$P_{CHP,t}^{e}/P_{CHP,t}^{h}$	electricity/heating power of CHP (kW)
P_{ES}^{ch}/P_{ES}^{dc}	charging/discharging power of ES (kW)
$P_{EC,t}^{C}/P_{EC,t}^{e}$	cooling/electric power of EC (kW)
$P_{GB,t}^{h}$	heating power of GB (kW)
$P_{PV}(t)$	PV output power (kW)
$P_{PV}^{R}(t)$	rated power of PV panel (kW/m^2)
P_{WT}	WT output power (kW)
P_{WT}^{R}	rated power of WT (kW)
$P_{WHB,t}^{h}$	heating power of WHB (kW)
$P_{i,load,t}^{e}/P_{i,load,t}^{h}/P_{i,load,t}^{c}$	power of load (kW)
$P_S(t)$	operation power of equipment s (kW)
$P_{i-j,t}^{e,P2P}/P_{i-j,t}^{h,P2P}$	P2P trading power of electricity/heating (kW) from IES i to j
$P_{j-i,t}^{e,P2P}/P_{j-i,t}^{h,P2P}$	P2P trading power of electricity/heating (kW) from IES j to i
s	set of equipment
t	time (h)

Superscripts

a	actual
c	cooling
ch	charge
CO	cooperative operation
dc	discharge
e	electricity
f	free
g	gas
h	heating
IO	independent operation
R	rated power
SP	subproblem

Greek symbols

$\delta_{grid}/\delta_{GB}^{h}$	carbon emission allowance for grid /GB (t/kWh)
$\delta_{CHP}^{e}/\delta_{CHP}^{h}$	carbon emission allowances of CHP (kg/kWh)
$\delta_{grid}/\delta_{GB}^{h}$	carbon emission allowance of grid/GB (kg/kWh)
$\gamma_{i-j}^{e,SP}/\gamma_{i-j}^{h,SP}$	Lagrange multiplier of subproblem
η_{PV}	energy conversion rate of PV
$\eta_{CHP}^{e}/\eta_{CHP}^{h}$	power/heat efficiency of CHP
$C_{i,om}$	maintenance cost of IES i (CNY)
$C_{i,CO2}$	carbon trading cost (CNY)
$C_{i,P2P}$	P2P trading cost (CNY/kWh)
$C_{i,net}$	network-crossing cost (CNY/kWh)
CE_t^f/CE_t^a	free/actual carbon emission (t)
COP_{AC}/COP_{EC}	coefficiency of performance of AC/EC
$E_{ES}^{\max}$	energy capacity of the ES (kWh)
$E_i^{sell}/E_i^{receive}$	sell/receive energy in P2P(kWh)
$E_{ES,t}/E_{ES,t-1}$	energy values of the ES at time t/t-1 (kWh)
M	set of IES
N	number of IES
$p_{buy,t}^{e}$	electricity price of grid (CNY/kWh)
$p_{sell,t}^{e}$	electricity price selling to the grid (CNY/kWh)
$p_{buy,t}^{h}$	heating price purchased from heating company (CNY/kWh)
$p_{\min,t}^{h}$	lower limit of heating price (CNY/kWh)
p_{net}^{e}/p_{net}^{h}	unit price of network-crossing charges (CNY/kWh)
$p_t^{e,buy}/p_t^{g,buy}$	electricity/gas price buying from outside (CNY/kWh)
p_{net}^{e}/p_{net}^{h}	unit price of network-crossing charges (CNY/kWh)
$P_{i,t}^{e,buy}/P_{i,t}^{g,buy}$	purchased power of electricity/gas (kW)
$P_S^{\max}/P_S^{\min}$	maximum/minimum power of the equipment s (kW)
$P_{i-j,t}^{e,\min}/P_{i-j,t}^{h,\min}$	minimum/maximum power of P2P trading for electricity between IES i-j (kW)
$P_{i-j,t}^{e,\max}/P_{i-j,t}^{h,\max}$	minimum/maximum power of P2P trading for heating between IES i-j (kW)
$\underline{\Delta P_S(t)}/\overline{\Delta P_S(t)}$	minimum/maximum limits of climbing speed of equipment s (kW)
Q_{gas}	calorific value of gas (kJ/m^3)
Soc_i	state of charge at time t
$SOC_{\min}/SOC_{\max}$	upper and lower limits of SOC
U_i/U_i^0	utility of IES i after/before cooperation
$v_R/v_{out}/v_{in}$	rated/cut-out/cut-in wind speeds (m/s)
v	wind speeds (m/s)
$V_{GB,t}^{g}, V_{CHP,t}^{g}$	gas power of GB/CHP (m^3)
$V_{P2P,i}$	contribution of IES I in P2P trading

strategy and scheduling model can cut the total cost by 10.7 %. Wang et al. (2023c) presented the distributed robust optimization scheduling model given the uncertainty of WT and PV within an IES and discussed the contribution of energy storage units. To cope with the uncertainty of renewable energy, Chen et al. (2024b) employed a robust optimization approach to address the IES network planning and operation problem. Based on the hybrid game strategy, Liang and Mu (2024) developed decision models for energy management operator, energy hub agent and user load aggregator to simulate the interactive transaction process of stakeholders. The above works are focused on the optimized operation of the individual IES and never cover the interactions among the IES. As the scale of renewable energy expands, the number of IESs will continue to rise. Peer-to-peer (P2P) energy trading across IESs has become a trend for future development. However, there are still relatively few studies on how to coordinate cooperative operation strategies within multiple IESs and rationally realize the optimal distribution of internal resources.

For optimized operation of multi-IESs, the approaches to investigating energy P2P trading and optimization problems contain centralized optimization method (Aminlou et al., 2024) and distributed optimization method (Aminlou et al., 2023). Regarding research on centralized trading methods, Huang et al. (2022) introduced the optimization framework considering the Stackelberg game for economic scheduling and energy-sharing strategy within IES clusters. The findings show that the joint energy-sharing alliance can enhance the benefits of members and facilitate the optimal utilization of energy resources. Hosseini and Ahmarinejad (2021) proposed a centralized two-layer collaborative operation model to address energy trading among multiple IESs. This model has demonstrated its effectiveness in reducing operating costs by approximately 5 %. A two-stage optimization framework was presented by Dorahaki et al. (2023) for the P2P electricity-heating energy trading in multi-carrier energy systems. Similarly, Rezaei and Ghasemi (2022) proposed a stochastic optimization strategy for multi-IES P2P electricity trading and the findings indicate that P2P trading reduces the load shedding of the system by 64 %. The above studies are all based on the precondition that there is a unified scheduling center among multi-IESs to design a unified operation and control strategy. In fact, when the cooperative alliance of multi-IESs engages in P2P energy transactions, the status of each IES is equal to each other. There still exists an internal game concerning economic interest (Wang et al., 2023a), which may lead to unfair distribution of benefits if a unified control center is required. Also, it is difficult to secure private information and increases communication costs. The distributed algorithm represented by the alternating direction method of multiplier (ADMM) gains wide concern related to IES multi-energy management (Cheng et al., 2024). Chen et al. (2024a) developed a cooperative operation strategy for multiple IESs with asymmetric Nash bargaining to determine the optimal electricity trading. However, the category is limited to electric energy. To develop the complementary capabilities of multi-energy systems, Babagheibi et al. (2023) proposed a P2P trading structure based on the ADMM to improve the social welfare value, reduce the operating costs, and consider the privacy-preserving requirements for the market clearing process. Duan et al. (2023) explored P2P electricity trading based on Nash bargaining considering the multiple uncertainties of renewable energy, energy price and energy demand. Li et al. (2024b) explored how P2P electricity trading could make the best use of complementary ability within the IES based on Nash bargaining. Similarly, Wang et al. (2024c) proposed an asymmetric Nash bargaining method to pursue the benefit distribution fairly in P2P trading alliances, but the above two works merely considered electrical energy sharing. As it continues with energy interconnections and energy markets, the P2P trading for electricity and heating energy between IESs remains to be explored. However, the above research still focused on the single electric energy flow. Although Xu and Yi (2023) proposed a standard Nash bargaining model for the electricity-heating sharing of a multi-IES and solved the model by the ADMM, this research did not discuss in-depth the benefit distribution of the alliance. Under the condition of complex IES structure and development, there is still a great gap in the research considering various energy types transactions.

When IESs are subject to different stakeholders operating cooperatively, it is of great significance to design a scientific energy trading and benefit distribution strategy. In existing studies, the leading methods of benefit distribution are the Stackelberg game (Li et al., 2023a), the Shapley value method (Gomes & Vale, 2024), and Nash bargaining (Meng et al., 2024). The standard Nash bargaining (SNB) is just to maximize the benefits of all members and distribute the profits on average. However, it is fairer to distribute the benefits according to each member's contribution. When the Shapley value method is employed to distribute benefits, there will be a "combination explosion" in the case of a huge participant group. It is imperative to develop an equitable and reasonable benefit distribution scheme to effectively settle the benefits of each member. To fill the gap, this study is devoted to proposing an improved Nash bargaining method by quantifying the contribution of IESs in P2P trading to determine the benefit distribution factor. Table 1 presents the differences between this study and other literature.

In summary, there are still several notable shortcomings in existing works. Firstly, the majority of current studies, exemplified by the works of Duan et al. (2023) and Kim et al. (2020), narrowly focus on electric energy sharing and neglect the potential for multi-energy sharing. This gap constrains the complementary potential among multiple IESs. Secondly, traditional benefit distribution methodologies, such as the Shapley value (Karimi et al., 2024), encounter difficulties in addressing the complexities of cooperative operations involving a larger number of IES participants. Efficient benefit distribution is paramount in fostering cooperation among IESs, yet non-cooperative game approaches, as exemplified by the work of Zhang and Fedorovich (2024), overly emphasize individual interests at the expense of global considerations. Furthermore, Nash bargaining ignores the consideration of individual contribution differences, thereby diminishing the incentive for IESs to engage in P2P energy trading.

To bridge the gap, this study presents an asymmetric Nash bargaining model for operation optimization and benefit distribution of multi-IESs considering P2P energy trading. This model aims to incentivize IESs to actively participate in P2P transactions and harness the full potential of multi-energy complementarity. The novelties of this paper are: (1) Diverging from the prevalent literature that solely emphasizes the optimization of individual IES, such as Tan et al. (2020); Zhao et al. (2023), this paper presents a novel P2P trading framework tailored for multi-IESs. Furthermore, it proposes a collaborative operation model that leverages Nash bargaining theory to transform the complex optimization problem into two subproblems: minimizing alliance costs and negotiating P2P trading payments. This approach enables a more comprehensive and efficient management of multi-IES interactions. (2) In contrast to previous studies, such as those conducted by Wang et al. (2022b) and Yang et al. (2024), which rely on traditional methods of distributing benefits, the paper introduces a concept of bargaining power factor to assess the participants' contribution to the P2P market, thereby ensuring a more equitable distribution of cooperative benefits. By adopting an asymmetric Nash bargaining model, the study not only achieves a fairer distribution but also significantly enhances the motivation of energy suppliers to engage in energy sharing. (3) Recognizing the importance of privacy protection, this study meticulously avoids the leakage of privacy information by integrating an ADMM algorithm for distributed solution. This algorithm not only ensures data privacy but also demonstrates superior convergence performance. The case study provides compelling evidence of the model's superiority, further underscoring the feasibility and rationality of the proposed strategy.

Table 1
Literature comparison.

Literature	Multi-IES	Category of energy sharing		Distributed optimization	Consider individual contribution differences	Distribution method of benefits/payments		
		Electricity	Heating			Shapley value	Standard Nash bargaining	Asymmetric Nash bargaining
(Jalving et al., 2023)	×	×	×	×	×	×	×	×
(Wang et al., 2022b)	×	×	×	×	×	√		
(Dorahaki et al., 2023)	√	√	√	×	×	×	×	×
(Fan et al., 2023)	√	√	√	√	×	×	×	×
(Chen et al., 2022)	√	√	×	√	×		√	
(Shi et al., 2023)	√	×	×	×	×	√		
(Xu et al., 2023)	√	√	×	√	×		√	
(Rajaei et al., 2021)	√	√	×	√	×	×	×	×
(Xu & Yi, 2023)	√	√	√	√	×		√	
(Zhang & Fedorovich, 2024)	√	√	×	√	×		√	
(Chen et al., 2024a)	√	√	×	√	√			√
This study	√	√	√	√	√			√

2. Material and methods

2.1. Multi-IESs framework

2.1.1. Structure of multi-IESs

Fig. 1 demonstrates the framework of P2P electricity-heating energy transactions for multi-IESs. The park acts as an integrated energy consumer and is supplied by the corresponding IES. The IESs are connected to the external energy network and purchase energy from the external energy network. When the IESs are organized into an alliance to operate collectively, the IESs within the alliance can share energy and conduct P2P transactions through the external network.

For one thing, natural gas belongs to the primary energy and is regulated by the authority. For another, cooling energy suffers from heavy loss during transmission and there is no cold transmission pipeline across the IESs. Hence, the P2P trading in this paper is only for electrical and heating energy. Transaction information and scheduling instructions are delivered through energy management centers. Within the alliance, P2P trading is possible while the supply power of a system runs inadequate and others have surpluses.

To carry out P2P trading, the following hypotheses should be complied with by each participant: (1) each integrated energy system operator is affiliated with different stakeholders, and both participants have the willingness to operate cooperatively; (2) each IES operator obeys the ``rational economic human being'' hypothesis, that is, pursuing their interests to be maximized;(3) The IES can operate grid-connected and also cooperate with other IESs. There are both external market and technical circumstances for cooperative operation between IESs. P2P transactions are conducted via the public energy network and the public network charges a service fee.

Fig. 2 presents the structure of the individual energy system. The IES incorporates combined heat and power (CHP), absorption chiller (AC), gas boiler (GB), electric chiller (EC), renewable energy (WT and PV), and energy storage systems. The demand consists of electricity, heating and

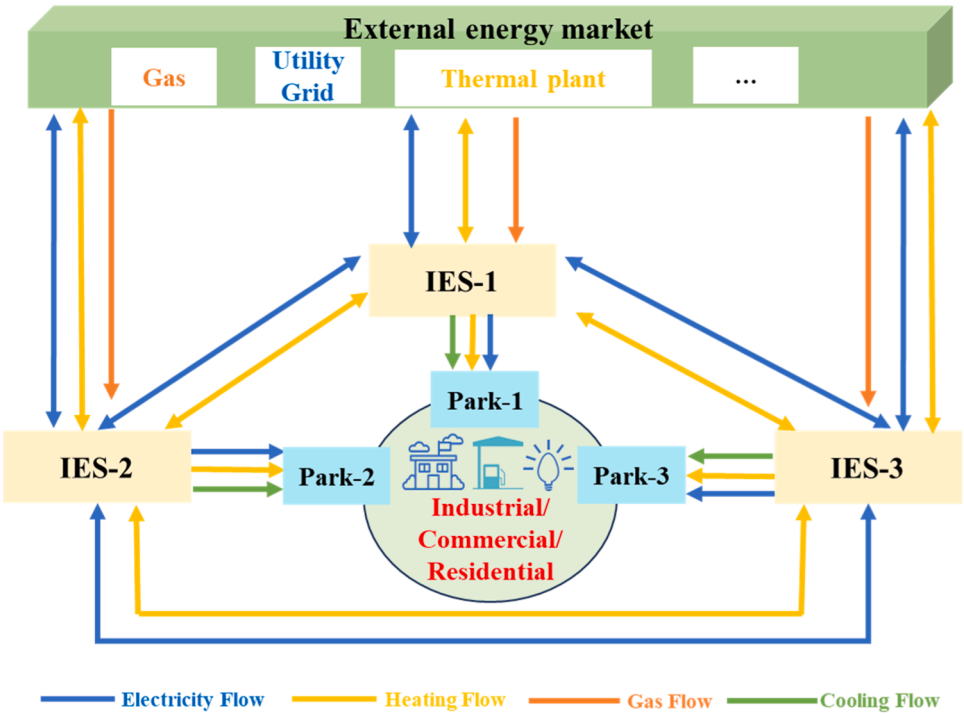


Fig. 1. P2P trading framework for multi-IES.

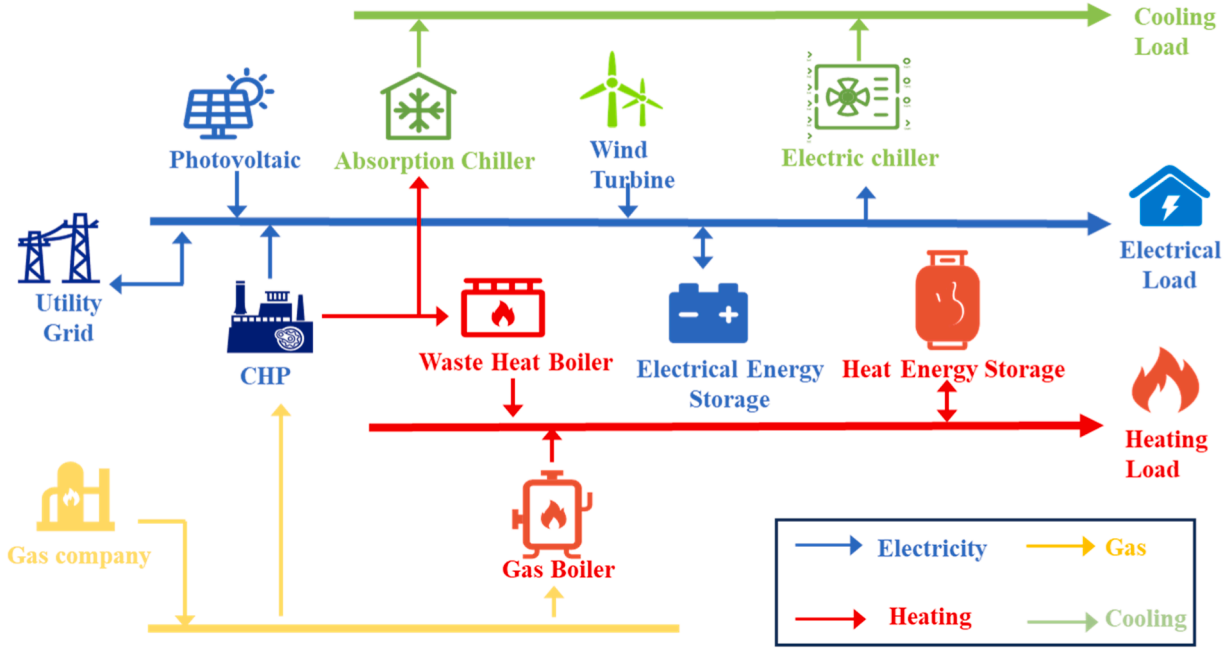


Fig. 2. Structure of IES.

cooling. The electricity demand is satisfied by the grid, electrical storage, WT and PV. For the heating demand, the heating generated by CHP is collected by the waste heat boiler (WHB) to cover the thermal demands; the shortfall is supplied by the GB, and the thermal storage can be used as thermal supplementary equipment. The cooling is mostly satisfied by AC and EC.

2.1.2. Models of equipment

Photovoltaics: The PV output depends on the rated power, intensity of solar radiation and energy conversion efficiency, which is modeled as follows (Ishaq & Dincer, 2024):

$$P_{PV}(t) = P_{PV}^{R}(t) A_{PV} \eta_{PV} \tag{1}$$

Wind Turbine: Wind power is determined by the rated power and wind speed, the model can be expressed (Karimi & Jadid, 2023; Song et al., 2024):

$$P_{WT}(t) = \begin{cases} 0 & v \le v_{in} or v > v_{out} \\ \dfrac{v - v_{in}}{v_R - v_{in}} P_{WT}^{R} & v_{in} < v \le v_R \\ P_{WT}^{R} & v_R < v \le v_{out} \end{cases} \tag{2}$$

Combined heat and Power: The CHP unit outputs electricity and heating energy simultaneously via feeding gas (Li et al., 2024a), and the heating produced by the CHP is captured by the WHB. The model is expressed as follows:

$$P_{CHP,t}^{e} = \eta_{CHP}^{e} V_{CHP,t}^{g} Q^{g} \tag{3}$$

$$P_{CHP,t}^{h} = \eta_{CHP}^{h} V_{CHP,t}^{g} Q^{g} \tag{4}$$

$$P_{WHB,t}^{h} = \eta_{WHB} P_{CHP,t}^{h} \tag{5}$$

Gas boiler: GB produces heating output via consuming gas, and the model is expressed as Karimi et al. (2024):

$$P_{GB,t}^{h} = \eta_{GB} V_{GB,t}^{g} Q^{g} \tag{6}$$

Absorption chiller: AC absorbs the thermal energy generated by the gas turbine and converts it into cooling energy. The model of AC is (Ren et al., 2024):

$$P_{AC,t}^{C} = P_{AC,t}^{h} COP_{AC} \tag{7}$$

Electric chiller: EC produces cooling energy to meet the cold demand by inputting electrical energy. The output model of EC can be expressed (Li et al., 2024b):

$$P_{EC,t}^{C} = P_{EC,t}^{e} COP_{EC} \tag{8}$$

Energy storage equipment: In this work, energy storage units include electrical energy storage (EES) and heating energy storage (HES). These units work on similar principles. A generalized expression is stated (Li et al., 2023b):

$$\begin{Bmatrix} E_{ES,t} = E_{ES,t-1} + \eta_{ES}^{ch} P_{ES,t}^{ch} - P_{ES,t}^{dc} \Big/ \eta_{ES}^{dc} \\ SOC_t = E_{ES,t} \Big/ E_{ES,t}^{max} \times 100\% \\ SOC_{min} \le SOC_t \le SOC_{max} \end{Bmatrix} \tag{9}$$

2.2. Collaborative operation model for IES alliance

2.2.1. Objective function

In an independent operation pattern, the operation costs C_i^{IO} of IES i consist of energy purchasing costs, carbon trading costs and equipment maintenance costs. Considering P2P trading under energy sharing, the cooperative operation cost C_i^{CO} will include a new item, that is P2P trading cost and network-crossing cost. Taking the IES i as an example, the objective functions in independent and cooperative operation modes are given (Fu et al., 2024):

$$C_i^{IO} = \min\left(C_{i,buy} + C_{i,om} + C_{i,CO2}\right) \tag{13}$$

$$C_i^{CO} = \min\left(C_i^{IO} + C_{i,P2P} + C_{i,net}\right) \tag{14}$$

When multiple IESs cooperate in an alliance, the objective of the alliance is to optimize the total operation costs (Fan et al., 2024). This is shown as follows:

$$\min \sum_{i=1}^{N} C_i^{CO} = \min \sum_{i=1}^{N} \left(C_{i,buy} + C_{i,om} + C_{i,CO2} + C_{i,P2P} + C_{i,net}\right) \tag{15}$$

The energy purchasing cost of IESs primarily consists of buying energy from the external grid and gas market, whose prices are determined by the superior energy market, and the cost of purchasing energy is given (Yang et al., 2023):

$$C_{i,buy} = \sum_{t=1}^{T} \left(p_t^{e,buy} P_{i,t}^{e,buy} + p_t^{g,buy} P_{i,t}^{g,buy} \right) \tag{16}$$

The equipment maintenance costs are below (Xiao et al., 2023):

$$C_{i,om} = \sum_{t=1}^{T} \sum_{s} \left(c_{i,s} P_{i,s,t} \right) \tag{17}$$

The network-crossing cost generated by P2P transactions should be borne by both parties. So, P2P trading cost and network-crossing cost are expressed as follows (Fan et al., 2024):

$$C_{i,P2P} = -\sum_{j \in l \backslash i} \sum_{t=1}^{T} \left(\lambda_{i-j,t}^{e,P2P} P_{i-j,t}^{e,P2P} + \lambda_{i-j,t}^{h,P2P} P_{i-j,t}^{h,P2P} \right) \tag{18}$$

$$C_{i,net} = \frac{1}{2} \sum_{j \in l \backslash i} \sum_{t=1}^{T} \left(p_{net}^{e} \left| P_{i-j,t}^{e,P2P} \right| + p_{net}^{h} \left| P_{i-j,t}^{h,P2P} \right| \right) \tag{19}$$

The carbon trading administration grants free carbon emission quotas to the IES according to the emission norms. When actual emissions exceed free quotas, they need to purchase the excess part from the carbon market. When actual emissions are less than the quota, the surplus part can be transacted for benefits. For the IES, the actual CO_2 emissions come from three main sources: electricity purchased from the grid, GB and CHP. So, the free carbon quota CE_t^f is illustrated as follows (Yang & Liu, 2024):

$$CE_t^f = \delta_{grid} P_{grid,t}^{e} + \left(\delta_{CHP}^{e} P_{CHP,t}^{e} + \delta_{CHP}^{h} P_{CHP,t}^{h} \right) + \delta_{GB}^{h} P_{GB,t}^{h} \tag{10}$$

Similarly, the actual carbon emission CE_t^a can be calculated by the formula (Fan et al., 2023):

$$CE_t^a = \tau_{grid} P_{grid,t}^{e} + \left(\tau_{CHP}^{e} P_{CHP,t}^{e} + \tau_{CHP}^{h} P_{CHP,t}^{h} \right) + \tau_{GB}^{h} P_{GB,t}^{h} \tag{11}$$

The carbon trading costs C_{CO2} are shown as Yang and Liu (2024):

$$C_{CO2} = \sum_{t} \pi \left(CE_t^a - CE_t^f \right) \tag{12}$$

2.2.2. Constraints

(1) Energy balance constraints (Salehimaleh et al., 2022)

$$P_{i,WT,t} + P_{i,PV,t} + P_{i,ES,t} + P_{i,GT,t}^{e} + P_{i,Grid,t} = P_{i,load,t}^{e} + P_{i,EC,t}^{e} + \sum_{j \in M/i} P_{i-j,t}^{e} \tag{20}$$

$$P_{i,CHP,t}^{h} + P_{i,GB,t}^{h} + P_{i,HES,t}^{h} = P_{i,load,t}^{h} + P_{i,AC,t}^{h} + \sum_{j \in M/i} P_{i-j,t}^{h} \tag{21}$$

$$P_{i,AC,t}^{C} + P_{i,EC,t}^{C} = P_{i,load,t}^{C} \tag{22}$$

(2) Equipment operation constraints

For any equipment, its operation power $P_S(t)$ at each period is limited by the minimum and maximum values (Xu et al., 2022), as well as the climbing speed constraints.

$$P_S^{\min} \le P_S(t) \le P_S^{\max} \tag{23}$$

$$\underline{\Delta P_S(t)} \le P_S(t) - P_S(t-1) \le \overline{\Delta P_S(t)} \tag{24}$$

(3) Price constraints on P2P trading

To avoid malicious markups, the price of internal P2P transactions is supposed to be in a definite range. To inspire IESs to participate in P2P transactions, the internal purchase and selling price should be better than the transaction price between the external market, that is (Zhang et al., 2022):

$$p_{sell,t}^{e} \le \lambda_{i-j,t}^{e} \le p_{buy,t}^{e} \tag{25}$$

$$p_{\min,t}^{h} \le \lambda_{i-j,t}^{h} \le p_{buy,t}^{h} \tag{26}$$

(4) Power constraints on P2P trading

The P2P trading power is required to remain within the limits of the network transmission power. Furthermore, the negotiated results of P2P trading volume should be consistent to guarantee a successful consensus (Fan et al., 2024).

$$P_{i-j,t}^{e,\min} \le P_{i-j,t}^{e,P2P} \le P_{i-j,t}^{e,\max} \tag{27}$$

$$P_{i-j,t}^{h,\min} \le P_{i-j,t}^{h,P2P} \le P_{i-j,t}^{h,\max} \tag{28}$$

$$P_{i-j,t}^{e} = -P_{j-i,t}^{e} \tag{29}$$

$$P_{i-j,t}^{h} = -P_{j-i,t}^{h} \tag{30}$$

2.3. Model transformations based on Nash bargaining

Nash bargaining, a category of cooperative games, is popularly applied to tackle multi-subject complicated optimization problems. It is an effective instrument for studying multisubject optimization. In this research, the IESs can obtain value-added benefits through cooperation to reduce costs and reach a binding agreement, which is in line with the requirements of the cooperative game. This paper applies Nash bargaining to determine the optimal alliance operation strategy and equilibrate the interests among stakeholders. By introducing the utility of IES i before cooperation and after cooperation, the standard Nash bargaining (SNB) model can be expressed (Wang et al., 2023a):

$$\begin{aligned} &\max \prod_{i=1}^{N} \left(U_i - U_i^0 \right) \\ &s.t. \quad U_i \ge U_i^0, \forall i \end{aligned} \tag{31}$$

Given the contribution gap between IESs to energy sharing, they have unequal voice and bargaining power in the negotiation. The bargaining power factorϕ_i is proposed here to show asymmetric bargaining, which implies the IESs possess distinct bargaining power in negotiations according to their respective voice rights. In this paper, the asymmetric Nash bargaining model is (Chen et al., 2024a):

$$\begin{aligned} &\max \prod_{i=1}^{N} \left(C_i^{IO} - C_i^{CO} \right)^{\phi_i} \\ &s.t. \quad C_i^{IO} \ge C_i^{CO}, \forall i \end{aligned} \tag{32}$$

In Eq. (32), the negotiation strategy is $\left\{ P_{i-j,t}^{e}, P_{i-j,t}^{h}, \lambda_{i-j,t}^{e}, \lambda_{i-j,t}^{h} \right\}$, which means P2P trading of the electricity power, heating power, electricity price, and heating price. The solution that satisfies Eq. (32) is the Nash bargaining equilibrium solution. Bringing Eqs. (13) and (14) into Eq. (32), it is found that there exists the multiplication of electricity (heating) and electricity price (heating price) in the Nash bargaining model, which is a typical multivariable coupling, non-convex nonlinear optimization issue and could hardly be solved directly. At the same time, the participants in the alliance come from different interest groups and fail to share their private information. In this regard, the Nash bargaining

method is equivalently transformed and decomposed into two sub-problems (Kim et al., 2020; Wang et al., 2024c), and the decomposition process is shown in the Appendix.

2.3.1. Subproblem 1 (SP1): minimizing the cooperative operation cost

A cooperative operation strategy for the alliance is to maximize the total benefits, which is equivalent to minimizing the alliance costs. The SP1 is defined as minimizing the cooperative operation cost (Chen et al., 2022):

$$\begin{aligned} &\min \sum_{i=1}^{N} \left(C_i^{IO} + C_{i,net}\right) \\ &s.t. \quad C_i^{IO} \geq C_i^{CO}, Eq(1) - Eq(30) \end{aligned} \tag{33}$$

In SP1, the objective function is to pursue the minimization of total operation costs, and the P2P trading costs offset each other within the alliance. Therefore, SP1 can only obtain the P2P trading power but not the P2P trading price of each IES. The decision variables obtained from SP1 are brought into Eq. (33), which represents the cost of cooperation $\left(C_i^{IO} + C_{i,net}\right)^*$.

2.3.2. Subproblem 2 (SP2): P2P trading payment cost bargaining

Once the electricity scale and the heating scale of P2P trading are determined, the prices of P2P trading must be identified. Given the contribution differences of the individual IES in energy sharing, they hold distinct bargaining power in the negotiation. The bargaining power factor is proposed here to show the asymmetric negotiation. The asymmetric Nash bargaining model is (Chen et al., 2024a):

$$\begin{aligned} &\max \prod_{i=1}^{N} \left[C_i^{IO,0} - \left(\left(C_i^{IO} + C_{i,net}\right)^* + C_{i,P2P}\right)\right]^{\phi_i} \\ &s.t. \quad C_i^{IO,0} \geq \left(C_i^{IO} + C_{i,net}\right)^* + C_{i,P2P} \end{aligned} \tag{34}$$

The bargaining power factor must satisfy the following characteristics: (1) The IES that provides both electrical and heating energy in P2P transaction contributes to the bargaining power. The more energy is shared, the stronger the bargaining power factor. (2) The bargaining power factor of the IES that does not engage in energy sharing is 0. (3) In P2P trading, the bargaining power of providing one unit of electricity and thermal energy is greater than that of receiving one unit of electricity and thermal energy. The bargaining power factor is as follows (Cui et al., 2021):

$$E_i^{sell} = \sum_{t=1}^{24} \left[\max\left(0, P_{i-j,t}^{e,P2P}\right) + \max\left(0, P_{i-j,t}^{h,P2P}\right) \Big/ \eta_{e-h}\right] \tag{35}$$

$$E_i^{receive} = -\sum_{t=1}^{24} \left[\min\left(0, P_{i-j,t}^{e,P2P}\right) + \min\left(0, P_{i-j,t}^{h,P2P}\right) \Big/ \eta_{e-h}\right] \tag{36}$$

$$V_{P2P,i} = \exp\left(E_i^{sell} \Big/ \max\left[E_i^{sell}, \forall i\right]\right) - \exp\left(-E_i^{receive} \Big/ \max\left[E_i^{receive}, \forall i\right]\right) \tag{37}$$

$$\phi_i = \frac{V_{P2P,i}}{\sum_i V_{P2P,i}} \tag{38}$$

To facilitate expression and conversion, the Eq. (34) is expressed in logarithmic form. Eventually, the energy trading payment bargaining strategy according to asymmetric Nash bargaining can be given (Chen et al., 20244a):

$$\begin{aligned} &\min \sum_{i=1}^{N} -\phi_i \ln\left[C_i^{IO,0} - \left(\left(C_i^{IO} + C_{i,net}\right)^* + C_{i,P2P}\right)\right] \\ &s.t. \quad Eq.(33); Eq.(35) - (38) \\ &\qquad C_i^{IO,0} \geq \left(C_i^{IO} + C_{i,net}\right)^* + C_{i,P2P} \end{aligned} \tag{39}$$

2.4. Model implementation solutions

To protect the privacy of participants in bargaining, the two sub-problems are solved in a distributed solution using ADMM. The ADMM algorithm is commonly employed to deal with optimization issues with separable variables due to its merits such as better convergence and simple formulation (Zhong et al., 2019).

2.4.1. Solutions to minimizing cooperation cost

Since SP1 contains the coupling variable of energy trading power between IESs, it needs to satisfy the sharing energy balance constraint. It should also realize that the energy purchased power is equal to the energy sold power within the alliance. Hence, an auxiliary variable is introduced to decouple it (Xu & Yi, 2023):

$$\begin{cases} P_{i-j,t}^{e} + P_{j-i,t}^{e} = 0 \\ P_{i-j,t}^{h} + P_{j-i,t}^{h} = 0 \end{cases} \tag{40}$$

In Eq. (40), $P_{i-j,t}^{e}$ and $P_{i-j,t}^{h}$ are the scale of electricity and heating energy that IES i expects to transact with IES j. While $\begin{cases} P_{i-j,t}^{e} = -P_{j-i,t}^{e} \\ P_{i-j,t}^{h} = -P_{j-i,t}^{h} \end{cases}$, the consensus of electricity and heating energy trading is established among alliances. After achieving the above targets, the solution of SP1 according to the ADMM algorithm is as follows:

(1) Create the augmented Lagrange function for SP 1 (Boyd, 2010).

$$\begin{aligned} L_i^{SP1} = {} & C_i^{IO} + C_{i,net} + \sum_{j=1 j\neq 1}^{N} \sum_{t=1}^{T} \gamma_{i-j}^{e,SP1} \left(P_{i-j,t}^{e} + P_{j-i,t}^{e}\right) + \sum_{j=1 j\neq 1}^{N} \sum_{t=1}^{T} \gamma_{i-j}^{h,SP1} \left(P_{i-j,t}^{h} + P_{j-i,t}^{h}\right) \\ & + \sum_{j=1 j\neq 1}^{N} \sum_{t=1}^{T} \frac{\rho_i^{e,SP1}}{2} \| P_{i-j,t}^{e} + P_{j-i,t}^{e} \|_2^2 + \sum_{j=1 j\neq 1}^{N} \sum_{t=1}^{T} \frac{\rho_i^{h,SP1}}{2} \| P_{i-j,t}^{h} + P_{j-i,t}^{h} \|_2^2 \end{aligned} \tag{41}$$

(2) Initialize the desired power of transaction for IES i, which is 0. Set the number of iterations k, convergence accuracy and Lagrange multiplier. Other IESs update their respective desired electricity and desired heating energy of transaction through local computation. The following steps are performed in every iteration:

IES i updates its decisions by the following equation $P_{i-j,t}^{e}(k+1)$, $P_{i-j,t}^{h}(k+1)$(Kim et al., 2020):

$$\begin{cases} P_{i-j,t}^{e}(k+1) = \text{argmin} L_i^{SP1}\left(\gamma_{i-j}^{e,SP1}(k), P_{i-j,t}^{e}(k), P_{i-j,t}^{e}(k)\right) \\ P_{i-j,t}^{h}(k+1) = \text{argmin} L_i^{SP1}\left(\gamma_{i-j}^{h,SP1}(k), P_{i-j,t}^{h}(k), P_{i-j,t}^{h}(k)\right) \end{cases} \tag{42}$$

The IES j receives the information $\left[P^{e}_{i-j,t}(k+1), P^{h}_{i-j,t}(k+1)\right]$ and updates its decision $\left[P^{e}_{j-i,t}(k+1), P^{h}_{j-i,t}(k+1)\right]$ according to the following equation (Duan et al., 2023):

$$\begin{cases} P^{e}_{j-i,t}(k+1) = \text{argmin} L_i^{SP1}\left(\gamma^{e,SP1}_{j-i}(k), P^{e}_{i-j,t}(k+1), P^{e}_{j-i,t}(k)\right) \\ P^{h}_{j-i,t}(k+1) = \text{argmin} L_i^{SP1}\left(\gamma^{h,SP1}_{j-i}(k), P^{h}_{i-j,t}(k+1), P^{h}_{j-i,t}(k)\right) \end{cases} \tag{43}$$

Repeat the above steps until all members have updated their trading electricity and heat in the current iteration.

(3) After the preceding iteration, update the Lagrange multipliers according to the following rule (Wang et al., 2022a):

$$\begin{cases} \gamma^{e,SP1}_{i-j}(k+1) = \gamma^{e,SP1}_{i-j}(k) + \rho_i^{e,SP1}\left(P^{e}_{i-j,t}(k) + P^{e}_{j-i,t}(k)\right) \\ \gamma^{h,SP1}_{i-j}(k+1) = \gamma^{h,SP1}_{i-j}(k) + \rho_i^{h,SP1}\left(P^{h}_{i-j,t}(k) + P^{h}_{j-i,t}(k)\right) \end{cases} \tag{44}$$

The IESs run repeatedly and only the power information is required during the iterations, while the penalty factors are updated until the convergence requirements are met. The convergence requirements are the original residuals and the dual residuals are expressed as follows (Wang et al., 2023b):

$$\begin{aligned} &\sum_{j=1,j\neq 1}^{N}\sum_{t=1}^{T} \| P^{e}_{i-j,t}(k) + P^{e}_{j-i,t}(k) \|_2^2 \le \sigma_1 \\ &\sum_{j=1,j\neq 1}^{N}\sum_{t=1}^{T} \| P^{h}_{i-j,t}(k) + P^{h}_{j-i,t}(k) \|_2^2 \le \sigma_1 \\ &\sum_{j=1,j\neq 1}^{N}\sum_{t=1}^{T} \| P^{e}_{i-j,t}(k+1) - P^{e}_{j-i,t}(k) \|_2^2 \le \xi_1 \\ &\sum_{j=1,j\neq 1}^{N}\sum_{t=1}^{T} \| P^{h}_{i-j,t}(k+1) - P^{h}_{j-i,t}(k) \|_2^2 \le \xi_1 \end{aligned} \tag{45}$$

2.4.2. Solutions to payment cost bargaining

SP2 contains coupling variables for the transaction prices across IESs, which are needed to ensure that the purchasing prices are equal to the selling prices. Hence, we introduce the auxiliary variables $\lambda^{e,*}_{j-i,t}$ and $\lambda^{h,*}_{j-i,t}$ to decouple the coupling variables as shown in the following equations (Wang et al., 2024b):

$$\begin{cases} \lambda^{e,*}_{j-i,t} = \lambda^{e}_{i-j,t} \\ \lambda^{h,*}_{j-i,t} = \lambda^{h}_{i-j,t} \end{cases} \tag{46}$$

When $\begin{cases} \lambda^{e}_{i-j,t} = \lambda^{e,*}_{j-i,t} \\ \lambda^{h}_{i-j,t} = \lambda^{h,*}_{j-i,t} \end{cases}$, it indicates that IES i and IES j have reached a consensus on the electricity price and heating price, SP2 is also performed by utilizing the ADMM algorithm in the following steps:

(1) Create the augmented Lagrange function for SP 2 (Boyd, 2010).

$$\begin{aligned} L^{SP2} = &-\phi_i \ln\left[C_i^{IO,0} - \left(\left(C_i^{IO} + C_{i,net}\right)^{*} + C_{i,P2P}\right)\right] + \sum_{j=1,j\neq 1}^{N}\sum_{t=1}^{T}\left[\gamma^{e,SP2}_{i-j}\left(\lambda^{e}_{i-j,t} - \lambda^{e,*}_{j-i,t}\right)\right. \\ &\left. + \gamma^{h,SP2}_{i-j}\left(\lambda^{h}_{i-j,t} - \lambda^{h,*}_{j-i,t}\right)\right] + \sum_{j=1,j\neq 1}^{N}\sum_{t=1}^{T}\left(\frac{\rho_i^{SP2}}{2}\| \lambda^{e}_{i-j,t} - \lambda^{e,*}_{j-i,t} \|_2^2 + \frac{\rho_i^{SP2}}{2}\| \lambda^{h}_{i-j,t} - \lambda^{h,*}_{j-i,t} \|_2^2\right) \end{aligned} \tag{47}$$

(2) Initialize the desired trading electricity price and heating price of the IES i to 0. Set the maximum number of iterations k, convergence accuracy, Lagrange multiplier and so on. Other IESs update their respective desired electrical price and desired heating price through local computation. The following update steps are performed in every iteration:

IES i updates its decisions $\lambda^{e}_{i-j,t}(k+1)$,$\lambda^{h}_{i-j,t}(k+1)$by the following equation (Kim et al., 2020):

$$\begin{cases} \lambda^{e}_{i-j,t}(k+1) = \text{argmin} L_i^{SP2}\left(\gamma^{e,SP2}_{i-j}(k), \lambda^{e}_{i-j,t}(k), \lambda^{e}_{i-j,t}(k)\right) \\ \lambda^{h}_{i-j,t}(k+1) = \text{argmin} L_i^{SP2}\left(\gamma^{h,SP2}_{i-j}(k), \lambda^{h}_{i-j,t}(k), \lambda^{h}_{i-j,t}(k)\right) \end{cases} \tag{48}$$

The IES j receives the information $\left[\lambda^{e}_{i-j,t}(k+1), \lambda^{h}_{i-j,t}(k+1)\right]$ and updates its decision $\left[\lambda^{e,*}_{j-i,t}(k+1), \lambda^{h,*}_{j-i,t}(k+1)\right]$ given the equation (Duan et al., 2023):

$$\begin{cases} \lambda^{e,*}_{j-i,t}(k+1) = \text{argmin} L_i^{SP2}\left(\gamma^{e,SP2}_{j-i}(k), \lambda^{e}_{i-j,t}(k+1), \lambda^{e,*}_{j-i,t}(k)\right) \\ \lambda^{h,*}_{j-i,t}(k+1) = \text{argmin} L_i^{SP2}\left(\gamma^{h,SP2}_{j-i}(k), \lambda^{h}_{i-j,t}(k+1), \lambda^{h,*}_{j-i,t}(k)\right) \end{cases} \tag{49}$$

Repeat the above steps until all members have updated their trading electricity price and heat price in the current iteration.

(3) After the preceding iteration, update the Lagrange multipliers according to the following rule (Wang et al., 2022a):

$$\begin{cases} \gamma^{e,SP2}_{i-j}(k+1) = \gamma^{e,SP2}_{i-j}(k) + \rho_i^{SP2}\left(\lambda^{e}_{i-j,t}(k) - \lambda^{e,*}_{j-i,t}(k)\right) \\ \gamma^{h,SP2}_{i-j}(k+1) = \gamma^{h,SP2}_{i-j}(k) + \rho_i^{SP2}\left(\lambda^{h}_{i-j,t}(k) - \lambda^{h,*}_{j-i,t}(k)\right) \end{cases} \tag{50}$$

The IESs run repeatedly and only the power information is required during the iterations, while the penalty factors are updated until the convergence requirements are met. The convergence requirements are the original residuals and dual residuals are expressed as follows (Wang et al., 2023b):

$$\begin{aligned} &\sum_{j=1,j\neq 1}^{N}\sum_{t=1}^{T} \| \lambda^{e}_{i-j,t}(k) - \lambda^{e,*}_{j-i,t}(k) \|_2^2 \le \sigma_2 \\ &\sum_{j=1,j\neq 1}^{N}\sum_{t=1}^{T} \| \lambda^{h}_{i-j,t}(k) - \lambda^{h,*}_{j-i,t}(k) \|_2^2 \le \sigma_2 \\ &\sum_{j=1,j\neq 1}^{N}\sum_{t=1}^{T} \| \lambda^{e}_{i-j,t}(k+1) - \lambda^{e,*}_{j-i,t}(k) \|_2^2 \le \xi_2 \\ &\sum_{j=1,j\neq 1}^{N}\sum_{t=1}^{T} \| \lambda^{h}_{i-j,t}(k+1) - \lambda^{h,*}_{j-i,t}(k) \|_2^2 \le \xi_2 \end{aligned} \tag{51}$$

The solution flows of the proposed model are shown in Fig. 3. The

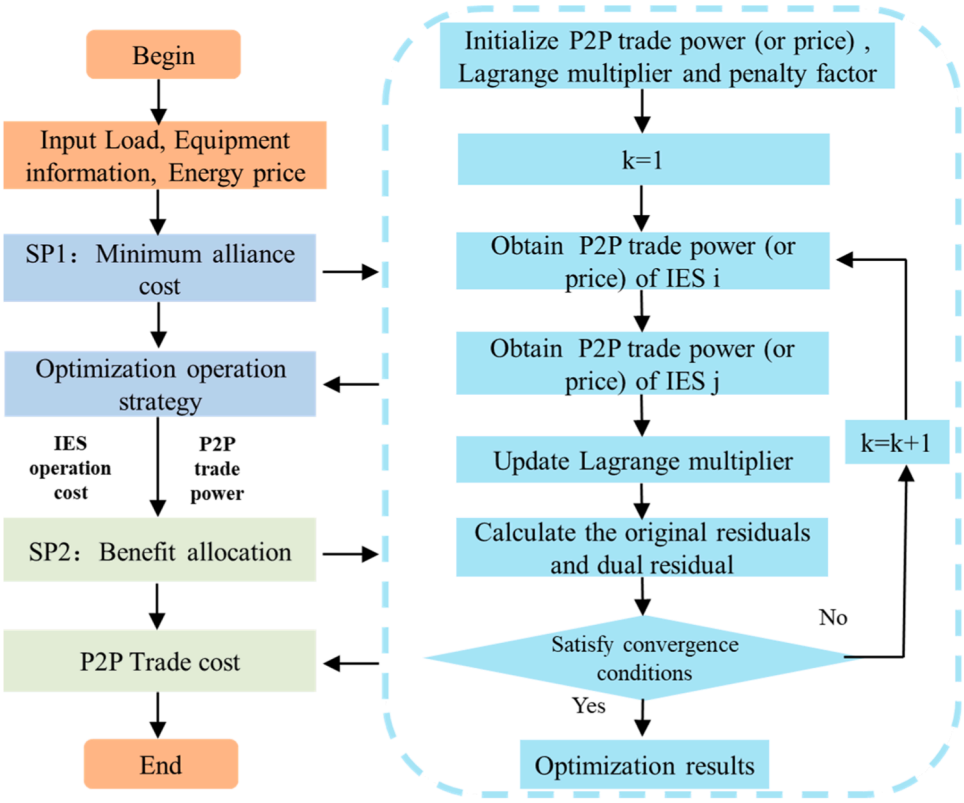


Fig. 3. Solution flows of the proposed model.

algorithm is performed on a computer with an Intel Core i5 processor and 16GB RAM. The proposed framework is run in MATLAB R2020a software.

3. Results and analysis

3.1. Basic data

In this study, a region containing three IESs is taken as an example for simulation. Each IES has a similar structure and Table 2 demonstrates the configuration information and economic parameters of each IES (Fan et al., 2023; Javadi et al., 2022). For few absent parameters of the devices are obtained through market inquiries. Fig. 4 gives the load and renewable energy data of IESs. The renewable energy sources in IES1 are wind power and PV, and in IES2 and IES3, the renewable energy is PV. The three IESs are geographically close to each other, so the purchasing price of electricity (TOU-E), heating (TOU-H) and gas are adopted from the local energy price, as shown in Fig. 5. The gas price is the average price published by the local energy agency, around 3.5 CNY/m^3 (Wang et al., 2023d). The network crossing charges for both electricity and heating energy in the region are set to 0.02 CNY/kWh. The maximum transmission limits of electrical power are 1200 kW, 600

Table 2
Technical and economic parameters of the equipment.

Equipment	Parameter	Value
CHP	Gas-heating efficiency/ Gas-electricity efficiency	0.45/0.35
	Installed capacity (kW)	1500/1500/1500
	Unit maintenance cost (CNY/kW)	0.025
GB	Efficiency	0.9
	Unit maintenance cost (CNY/kW)	0.012
	Installed capacity (kW)	800/800/600
EC	Coefficiency of performance	3.5
	Unit maintenance cost (CNY/kW)	0.012
	Installed capacity (kW)	800/800/600
AC	Coefficiency of performance	1.2
	Unit maintenance cost (CNY/kW)	0.016
	Installed capacity (kW)	600/600/600
EES	Efficiency	0.95
	State of charge	0.2–0.9
	Installed capacity (kWh)	1500
	Unit maintenance cost (CNY/kW)	0.009
HES	Efficiency	0.9
	State of charge	0.2–0.8
	Installed capacity (kWh)	2000
	Unit maintenance cost (CNY/kW)	0.005
Grid	Maximum purchasing power (kW)	2000
Heating network	Maximum purchasing power (kW)	1000

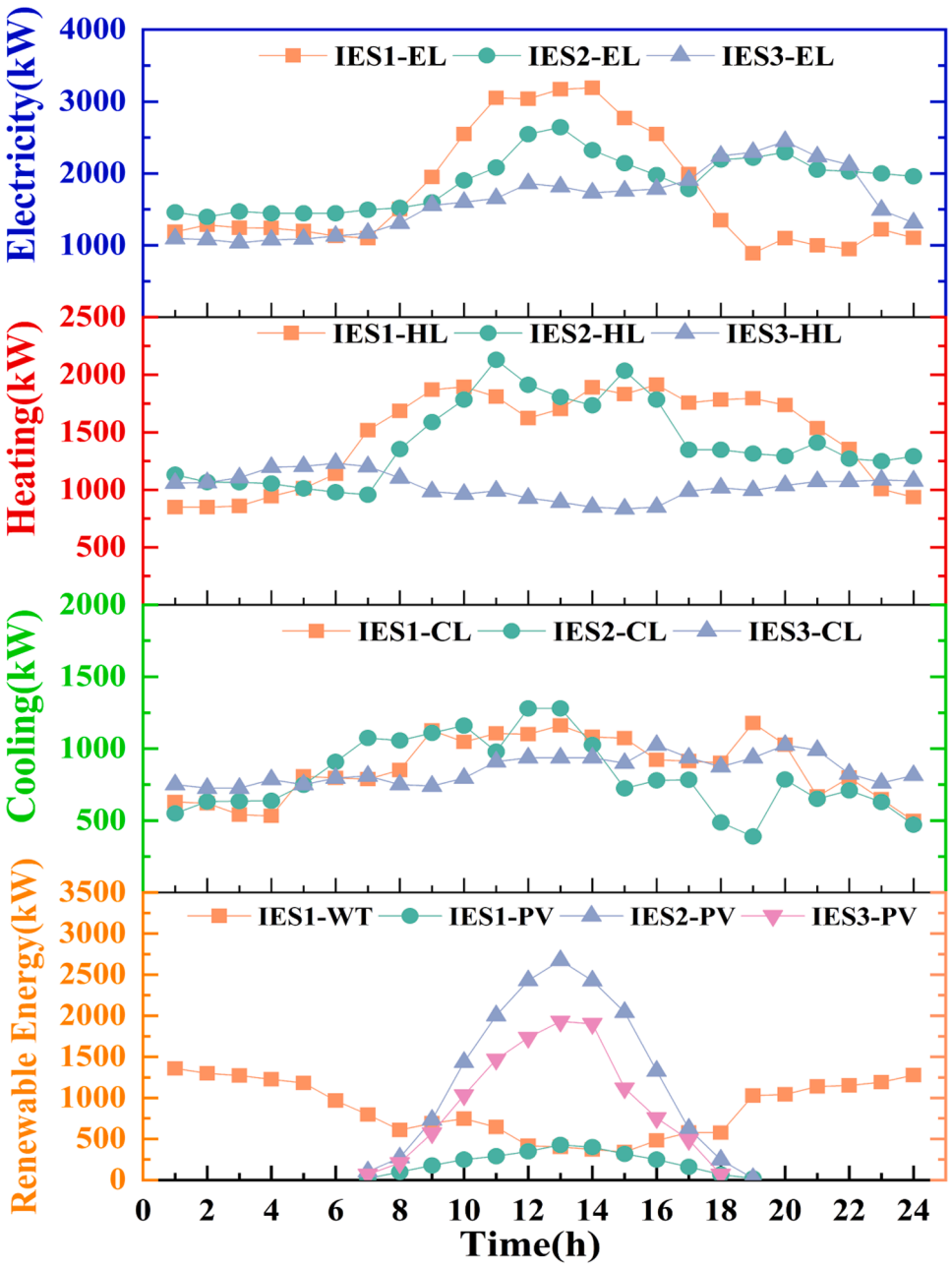


Fig. 4. Load profile and renewable energy output.

kW and 1000 kW for IES1-2, IES1-3 and IES2-3, respectively, and 600 kW, 400 kW and 600 kW for thermal power when P2P transactions are conducted. The initial value of the Lagrange multiplier and penalty factor is set as 0 and 0.005.

3.2. Algorithm convergence analysis

Fig. 6 shows the convergence process of SP1. During the initial stage of calculation, each IES ignores the coupling constraints in the alliance and the cost fluctuates greatly. With the increasing number of iterations, the information interaction between the interconnected systems is adequate, the Lagrange multiplier term gradually accumulates the constraint coupling variables asymptotically, and the trend of the cost changes slows down and finally converges after 12 iterations.

Fig. 7 depicts the convergence process of SP2. SP2 achieves convergence after 8 iterations. The changing trend of the transaction cost for each IES is varied. The transaction costs of IES1 tend to decline at large, while the counterpart of IES3 is the opposite. The slight fluctuation of the cost during the iteration process vividly reflects the negotiation progress of each participant concerning transaction price. The sum of alliance transaction costs approaches 0, which satisfies the internal equilibrium requirement for energy trading within the alliance.

3.3. Analysis of P2P transaction optimization

3.3.1. Results of P2P trading

Energy sharing among IESs is the core of achieving multi-energy complementarity and improving system energy efficiency. The ability to supply energy is no longer solely limited to the affordability of individual supply, and the IES that suffers from a lack of energy could acquire supplements from the IESs that are abundant in energy.

Fig. 8 displays the process of the energy interactions. During the night, when the load is low, IES1 has more wind power generation, so it transmits the excess renewable energy to IES2 and IES3 during this time.

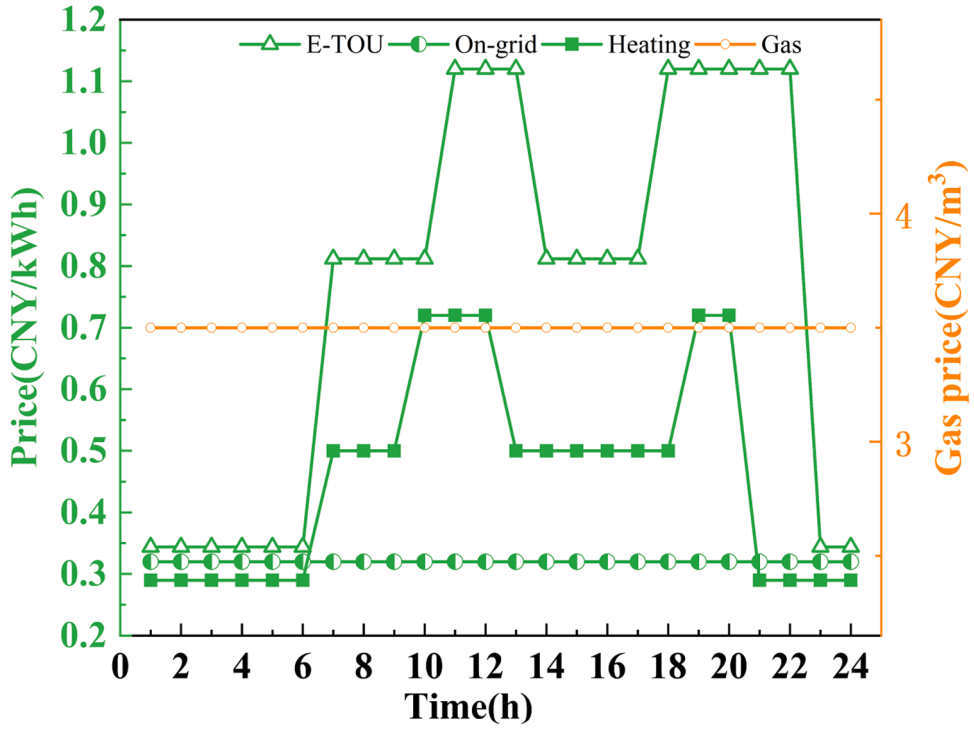


Fig. 5. Energy prices.

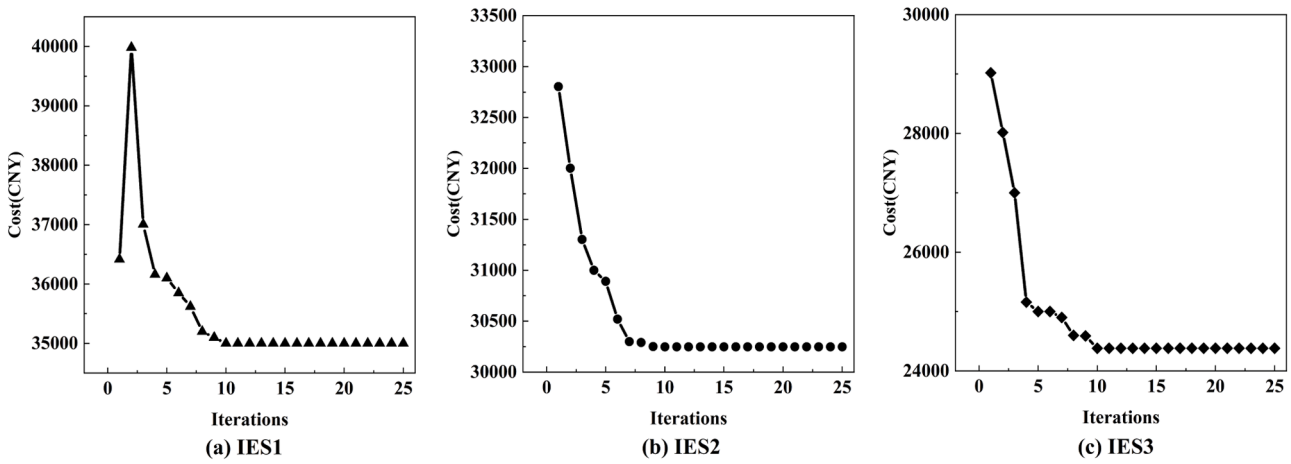


Fig. 6. Convergence process of SP1.

During the daytime from 10:00 to 15:00, the supply-demand matching is opposite to that of the nighttime, and IES1 purchases energy from IES2 and IES3. Meanwhile, the CHP in IES1 operates at full capacity and generates surplus heating that can be sold to other IESs. The thermal demand of IES2 and IES3 during the period of 10:00–15:00 is also at a high level. In other hours, the output of equipment for each IES can match its demand, and there is a small amount of thermal energy interaction. Moreover, the P2P transaction power of electrical and thermal energy is symmetric about the coordinate axis, which also verifies that the obtained results satisfy the power coupling constraints.

To further investigate the influence of P2P transaction on the performance of operation. Fig. 9 presents the consumption of renewable energy in independent operation mode and cooperative operation mode. Fig. 10 displays the change of electricity purchased from external market of each IES. The introduction of P2P trading promotes the ability of renewable energy consumption, and the utilization rate of renewable energy in each IES is increased by 28 %, 31 %, and 12 %, respectively. Also, energy sharing within the alliance reduces the dependence on the external energy network, and the alliance accumulatively reduces the amount of electricity purchased from the external by 12,685 kWh.

3.3.2. *Analysis of energy balance*

Take the IES1 as an example to analyze its energy scheduling strategy. As shown in Fig. 11, all kinds of energy in IES1 realize the balance of supply and demand. Concerning the electric energy scheduling scheme, renewable energy shares the highest priority in the energy scheduling scheme owing to its low cost, and no carbon emission. When wind power and photovoltaic cannot satisfy the demand, the supply is increased by CHP, electrical energy storage and P2P trading. The CHP in IES1 generates electricity during the period of 8:00–19:00, which reduces the purchasing power during the peak and flat periods of the tariff. Only a few electricity comes from the external grid. The operation behavior of energy storage is dynamically and flexibly adjusted according to the implication of time-of-use tariffs. Charging is carried out at night when

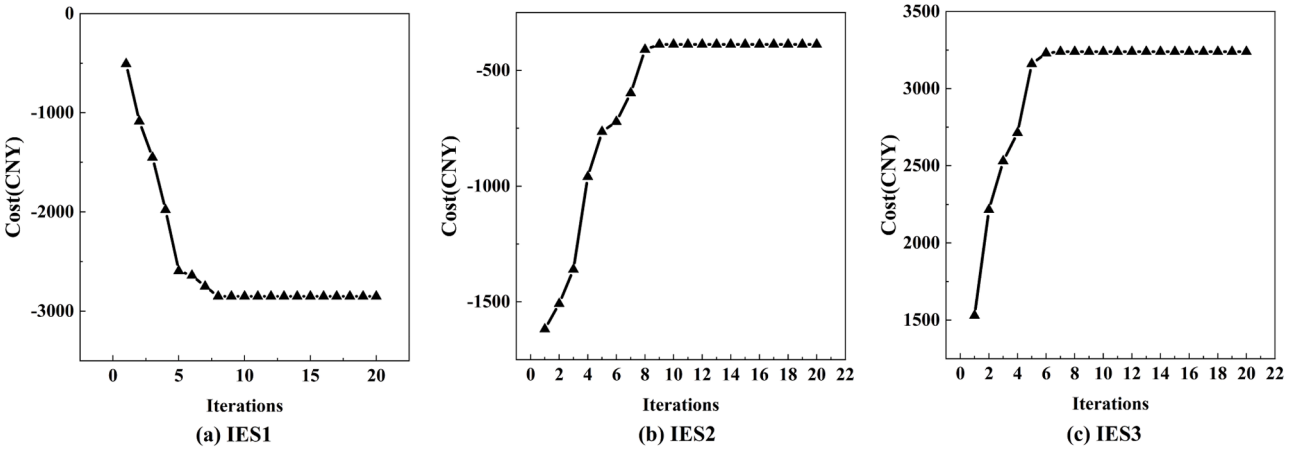


Fig. 7. Convergence process of SP2.

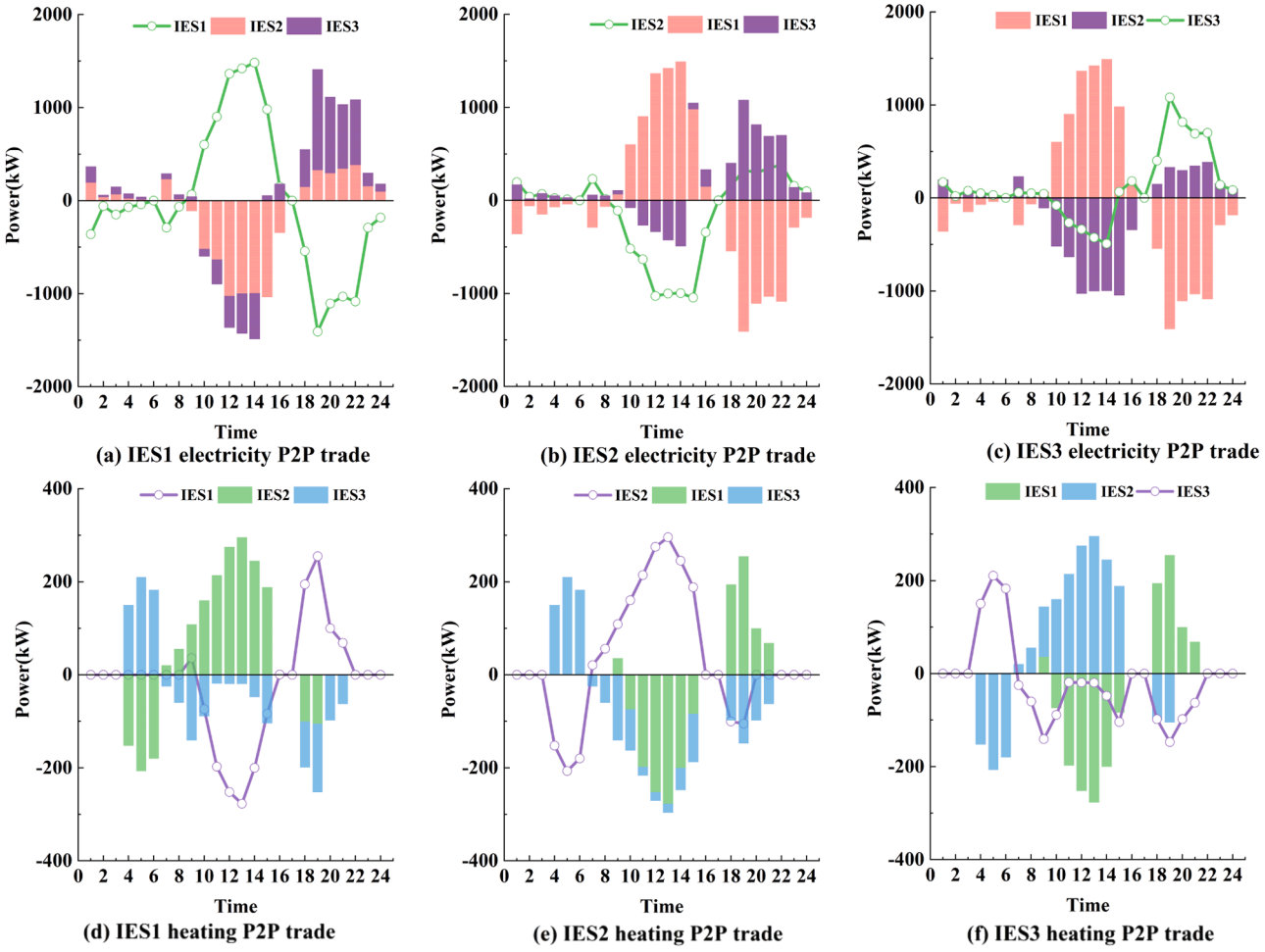


Fig. 8. P2P transactions of electricity and heating energy among IESs.

the electricity price is cheaper, and discharging is carried out at peak hours of power price. Thus, it realizes the transfer of electricity in the time dimension and makes full use of its peak-shaving and valley-filling functions. In addition, the P2P transaction provides an additional alternative for IES1 and cuts down the total operation cost.

In the heating energy scheduling strategy, thermal energy is supplied primarily by CHP and GB, and it is combined with thermal storage devices and interaction with other IESs to cover the thermal energy demand. As the thermal energy demand is largely satisfied by CHP, when the electrical power of CHP is reduced during the low price of electricity, the gap of thermal energy is filled by GB and thermal storage devices. Heating energy interacting with other IESs is utilized when the supply in IES1 is still Insufficient. For the cooling scheduling scheme, the cooling demand is small and is met by EC for most of the period, while AC is used to make up the shortfall in periods from 9:00 to 20:00.

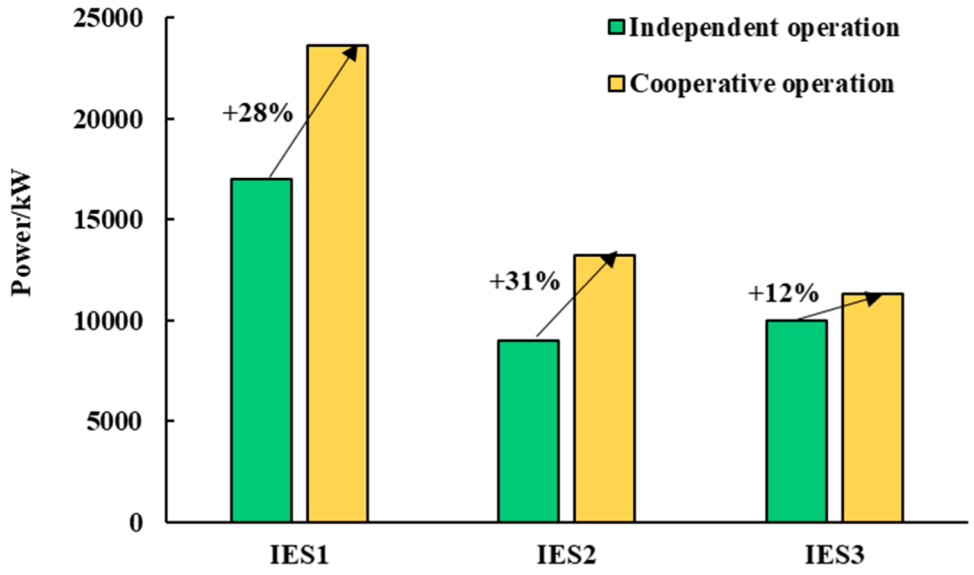


Fig. 9. P2P transactions of electricity and heating energy among IESs.

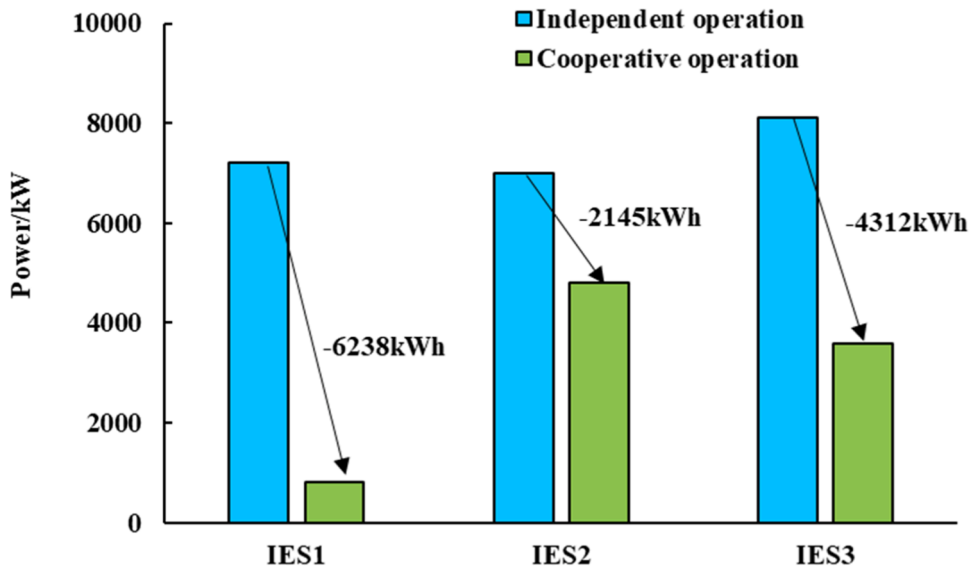


Fig. 10. P2P transactions of electricity and heating energy among IESs.

3.4. Analysis of benefit distribution

To test the merits of the proposed model, three scenarios are carried out:

Scenario 1 (S1): Every IES operates independently and the system solely trades with the external energy grid. That is the cooperation breakdown point;
Scenario 2 (S2): Each IES operates cooperatively. The system conducts P2P electricity-heating transactions and the system is performed by the standard Nash bargaining method;
Scenario 3 (S3): Each IES operates cooperatively, and the system carries out electricity-heating P2P transactions. The system is performed by the asymmetric Nash bargaining model, which is proposed in this paper.

Fig. 12 shows the trading prices of electricity and heating energy for each IES. The electricity transaction prices are cheaper than the price of the grid, but the prices are higher compared with the electricity price sold to the grid. Also, the thermal energy trading price is at a relatively low interval. As a consequence, when there is an insufficient supply of electricity or heating energy within the IES, IES within the alliance will prioritize the P2P trading with the energy-rich IES which offers a lower energy price, and the IES which sells the energy will benefit from it. Consequently, the operating costs of the IESs are effectively reduced.

With the distributed solution of the alliance cost minimization and payment bargaining subproblems, the operation cost of each IES is finally determined, as shown in Table 3. In S1, the IES operates without cooperation, making information enclosed owing to the absence of communication with other IESs. Its operation cost becomes the highest. Compared to S1, the operation cost of each IES is reduced in S2 and S3. The benefits of the alliance increase by 8,605CNY. The positive transaction cost of IES3 indicates that more energy is purchased from other IESs. The negative transaction cost of IES1 and IES2 indicates that the above-mentioned energy systems gain the benefit of selling more energy in P2P transaction. When benefit allocation in S2 is performed based on the standard Nash negotiation method, the energy provider and receiver are considered to contribute equally in P2P transaction, and therefore

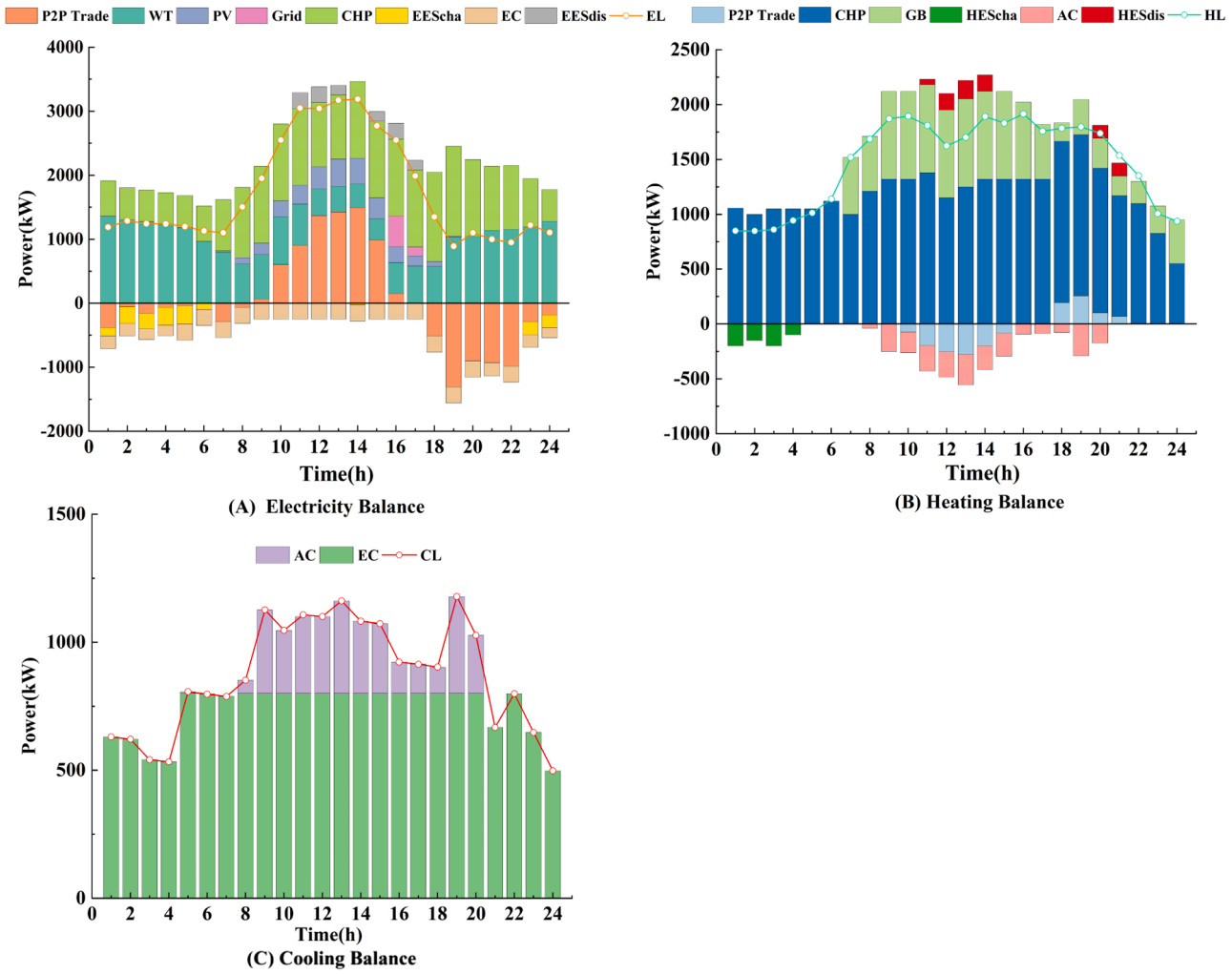


Fig. 11. Energy balance of IES1.

the Nash negotiation is a symmetric mechanism. The distributed benefit of each IES is essentially equal, which is about 2868.33CNY. When the scale of energy contribution of each IES in the cooperation is not equal, it is obvious that this method is unfair. It fails to reflect the bargaining characteristics of different IESs and can hardly create a competitive environment.

In S3, the energy suppliers are recognized to contribute more than the receivers in the P2P transaction as shown in Fig. 13. Therefore, the Nash bargaining is asymmetric, which means that the more contributions, the stronger the bargaining power. Upon cooperation among IESs, each IES can reduce the transaction with external energy grids and increase the profit by sharing electricity and heating energy. In S3, the contribution of electricity and heating energy quantified by the proposed method is regarded as the weights for the benefit distribution. The benefits obtained by each IES are appropriately evaluated by the weights. IES 1 contributes more electricity and heating energy than IES 2 and IES 3. IES 3 gives the least contribution. The final aggregate weight is determined by combining electricity and heating. The greater the contribution potential, the higher the aggregated weight, and the more benefits received in the end. In contrast to the SNB, the benefit distribution of the proposed method allows IESs to obtain a more equal return in line with their specific contribution.

3.5. Discussion and comparison

3.5.1. Effect of network-crossing charges

The network-crossing charges determine the scale of P2P transactions to a large extent. Fig. 14 presents the trend of P2P transaction amount with the change of the network-crossing charges price. With the prices increasing, the total amount of P2P transactions is gradually reduced. When the price is in the range of 0–0.1, the total amount of P2P transactions changes by a greater margin, which indicates that the volume of transactions in this interval is more sensitive to the price. When the price is in the range of 0.1–0.15, the sensitivity of the P2P transaction size is weakened. As the price continues to grow, P2P transactions are terminated. This indicates that the cooperation between the IESs breaks down due to paying excessively high network-crossing fees.

Compared to the independent operation cost, the network-crossing charges increase the cooperative operation cost, but P2P transaction reduces the cooperative operation cost. In this paper, the simulation parameters are set lower, so the total running cost is decreasing. When the added cost of network-crossing charges is equal to the saved cost via P2P transaction, the cooperative alliance is hitting critical mass. Fig. 15 demonstrates the sensitivity of the IES operation cost to the network-crossing charges. The cost of each IES increases as the network-crossing charges increase in the cooperative pattern. In particular, the amount of change is more pronounced for IES1. The main reason is that IES1 has the largest P2P transaction volume so it is most remarkably affected.

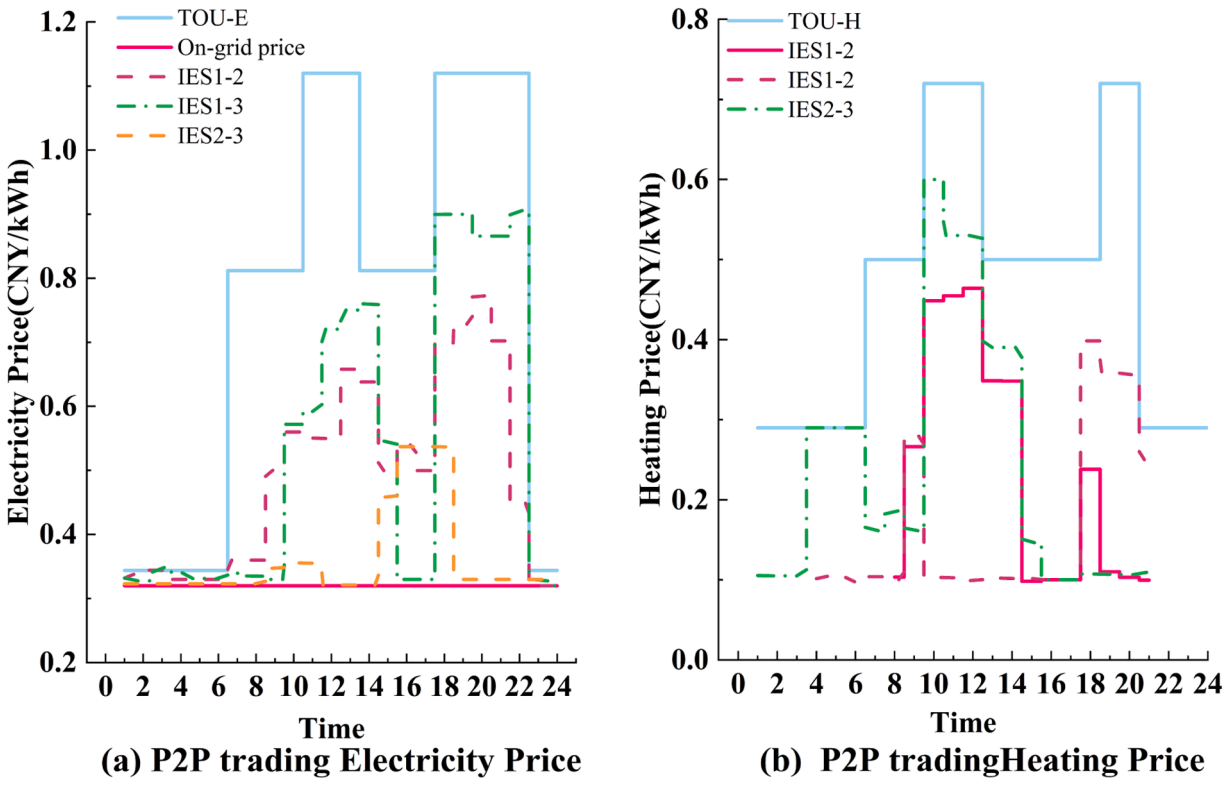


Fig. 12. Energy trading prices in S3.

Table 3
IES operation cost in different scenarios (unit: CNY).

IES	S1	S2					S3				
		Basic cost	Barging power	Payment cost	Final cost	Benefit increase	Basic cost	Barging power	Payment cost	Final cost	Benefit increase
1	36,415.27	35,001.53	0.333	−1454.59	33,546.94	2868.33	35,001.53	2.350	−2851.35	32,150.18	4265.09
2	32,803.12	30,247.42	0.333	−312.633	29,934.79	2868.33	30,247.42	1.622	−387.99	29,859.43	2943.69
3	29,018.97	24,383.41	0.333	1767.23	26,150.64	2868.33	24,383.41	0.769	3239.33	27,622.74	1396.23

Remarkably, in this study, P2P transactions are conducted through an external energy network for the transmission. As such, the price of the network-crossing charges is set at a low level. In the actual project, when the participating entities construct their own energy networks for P2P transactions, the investment and construction costs of the energy networks will be higher. All investment costs will be transferred to the P2P transaction price; in that case, the P2P transaction price will be higher than the price in this work.

3.5.2. *Effect of operation strategies*

Table 4 shows the total cost and required information under different operation and solution strategies. Among them, S1 is the independent operation mode, S2 is the cooperative operation but with a centralized solution method, and S3 is the cooperative operation but with a distributed solution, that is the scenario of this paper. The level of shared information requirement for IES is 0–2, where 0 is no sharing of information, 2 is required for full sharing of information, and 1 is partial sharing of information.

In general, although the model convergence time under the cooperative operation mode (S2 and S3) is increased compared with that under the independent operation strategy (S1), the economic benefits of cooperative operation are significantly improved. The value of the benefit improvement is 8909.65CNY and 8495.35CNY, respectively. The convergence time of the distributed solution method in this study is shortened from 537 s to 376 s than the centralized solution, and the solution efficiency is improved by nearly 30 %.

The difference in total operation cost between distributed and centralized solutions is small. The above results indicate that distributed solution shows a significant improvement in economic performance compared with the independent mode. That demonstrates the effectiveness of the distributed solution algorithm. The centralized approach requires the internal information of all the IESs, and each IES needs to upload its internal decision variables and parameters, while the distributed approach only requires each IES to exchange information about its P2P transaction power, which requires less sharing information and has better privacy protection. In summary, the distributed solution algorithm in this work can realize the system's demand in an efficient way with fewer information-sharing requirements and good convergence performance. When more stakeholders are participating in the operation of the multi-IESs, it will be more difficult to share private information of the IESs. The distributed solution to multi-IESs, which is proposed in this work, shows great potential with the development of the diversity of stakeholders in the IES.

3.5.3. *Comparison with literature*

To further discuss the rationality of the proposed benefit distribution method, the results of this paper are compared with existing literature. Among them, Xu and Yi (2023) merely consider the P2P

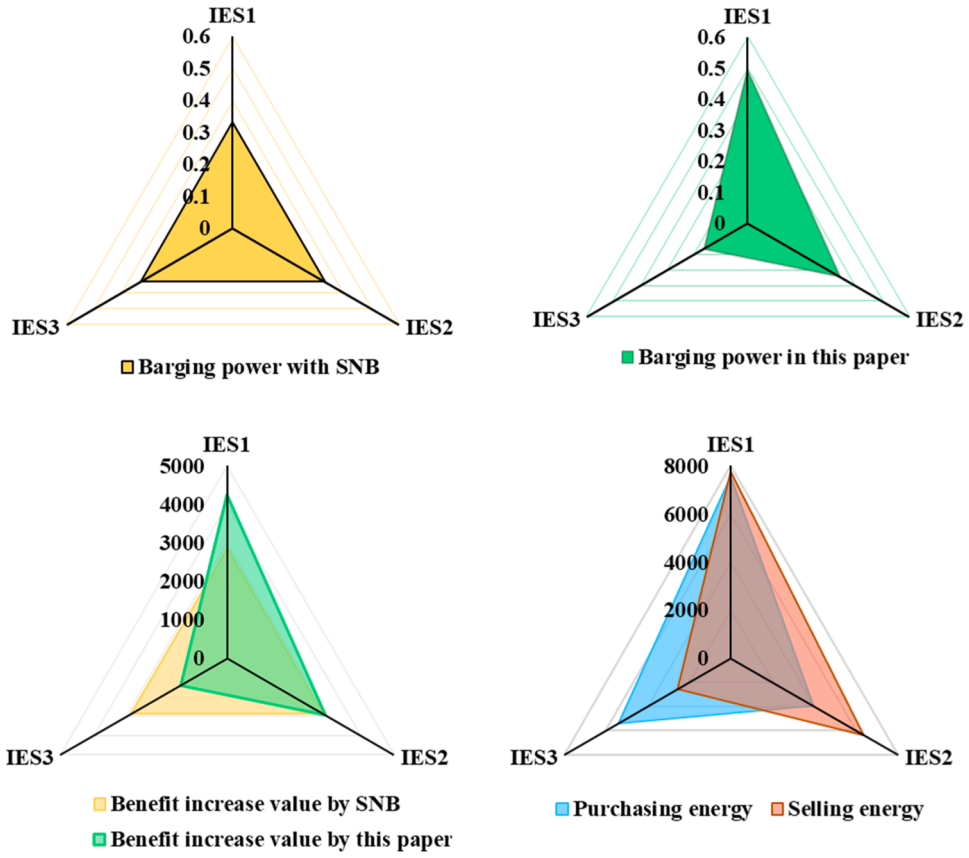


Fig. 13. Bargaining power and benefit-sharing values for S2 and S3.

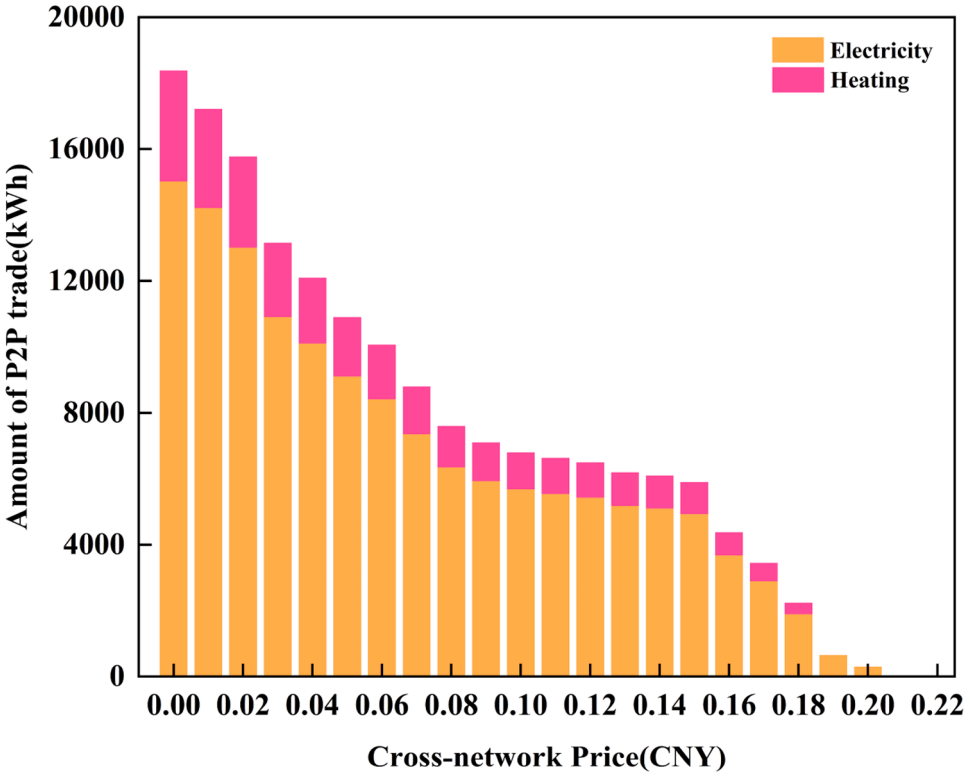


Fig. 14. Sensitivity of transaction volume to network-crossing charges price.

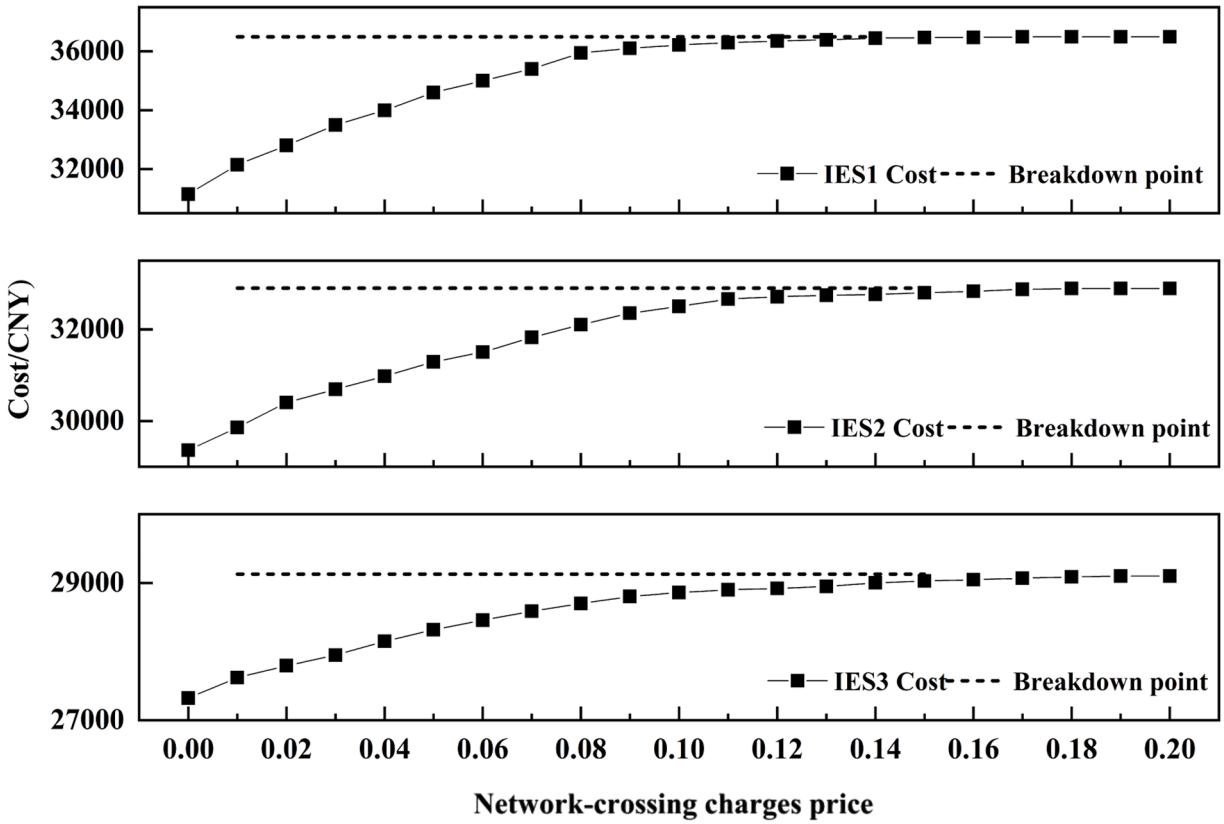


Fig. 15. Sensitivity of each IES operation cost to network-crossing charges.

Table 4
Total cost and required information under different solutions.

Scenarios	Total cost (CNY)	Iterations number	Convergence time (S)	Required sharing information
S1	98,237.36	5	86	0
S2	89,627.71	17	537	2
S3	89,632.36	12+8	376	1

Table 5
Comparison of operation cost with other literature (unit: CNY).

IES	Independent operation	Coordinated operation		
		(Xu & Yi, 2023)	(Wang et al., 2022a)	This paper
IES 1	36,415.27	33,546.94	32,871.43	32,150.18
IES 2	32,803.12	29,934.79	30,025.87	29,859.43
IES 3	29,018.97	26,150.64	28,003.10	27,622.74

electricity-heating trading between IESs and fail to consider the individual contribution gap; Wang et al. (2022a) consider the individual contribution gap in the cooperative alliance but merely performs the P2P electricity trading. The benefit distribution results are shown in Table 5.

Compared with the independent operation, the cooperative operation of IESs performing P2P energy trading could reduce the internal operation cost. However, Xu and Yi (2023) focus on the electric-thermal energy interactions between IESs and they fail to discuss the contribution gap of each participant in the P2P market. Therefore, the reduced cost of each IES is 2868.33CNY when the standard Nash bargaining model is used for benefit distribution. In the process of P2P electricity trading conducted by Wang et al. (2022a), the cooperative benefit allocation is conducted by quantifying the contribution of the electricity receiver and supplier. It is obvious that IES 1, as the most important electricity supplier, assigns the most benefits (about 4268.24 CNY). However, Wang et al. (2022a) did not consider the heating energy trading, so the total operation cost of the system is higher than that of this paper. In contrast, this paper fully considers the P2P electricity-heating transaction and individual contribution gap, which effectively reduces the overall operation cost of the alliance. On the other hand, the appropriate benefit distribution stimulates each IES to strengthen its bargaining power to obtain more rewards. In this way, the IES which contains abundant resources is greatly inspired to engage in the P2P market.

In practice, if only a single energy is taken into account in P2P trading, it is impossible to utilize the best operation of the alliance. Even though the traditional bargaining approach can still elevate the performance of multi-IESs, it fails to assign the benefits of cooperation effectively. The proposed model adopts the asymmetric Nash bargaining strategy to capture more benefits by giving the energy supplier greater bargaining strength. For the energy receiver of the P2P transaction, although the bargaining power is weaker than that of the energy supplier, the internal energy price of the alliance is still cheaper than the external energy price. Hence, the proposed model can realize a win-win solution for all participants in P2P transactions. The proposed model is more practicable in its application and can inspire the IES to positively contribute to P2P transactions.

3.6. Limitations and challenges of P2P energy trading

Although the findings of this study suggest that P2P energy trading can enhance system economics and that a distributed optimization approach can improve the efficiency of model solutions. However, P2P

energy transactions among integrated energy systems face several challenges and difficulties, including the following:

(1) Inadequate policy and regulatory frameworks: P2P energy trading across integrated energy systems involves multiple administrative regions and regulatory authorities, and it is difficult to form a harmonized regulatory framework because energy policies and regulations may be inconsistent across different regions. This regulatory incoherence may lead to unclear trading mechanisms, increasing the difficulty and uncertainty of trading. (2) Technical standards and interconnection issues: P2P energy transactions of multi-integrated energy systems involve different types of energy (e.g., electricity, heat, natural gas, etc.), and the technical standards and equipment used in different regions may be different, leading to technical barriers to interconnection between systems.(3) Difficulty in matching energy demand and supply: It is not easy to match the energy supply and demand of a multi-integrated energy system; the energy demand characteristics of different parks are different, with obvious seasonal fluctuations, making it difficult to realize real-time and accurate energy sharing. Inaccurate demand forecasting may lead to energy waste or shortage. (4) Infrastructure investment and maintenance: P2P energy transactions among integrated energy systems require a large amount of infrastructure investment, including energy transmission and distribution networks, smart metering equipment, and so on. These infrastructures require high construction costs, long operation periods, and significant challenges in post-production maintenance and management.

These challenges require sustained efforts in policy formulation, technological research and development, and market mechanism construction to facilitate the smooth implementation of P2P energy transactions among integrated energy systems.

4. Conclusion

P2P energy trading between integrated energy systems can effectively improve resource allocation efficiency and achieve supply-demand balance. How to coordinate the operation of multiple integrated energy systems and distribute cooperative benefits is a hot topic. Therefore, this paper introduces the asymmetric Nash bargaining method and establishes a coordinated operation model of multiple integrated energy systems. This model is equivalent to the minimum operating cost subproblem and payment negotiation subproblem. The proposed model is addressed by a distributed solution algorithm. The main conclusions are as follows:

(1) The proposed coordinated operation model can consider both the individual economic benefits and the global economics. Through a coordinated operation strategy, the benefits of each member are improved by 11.7 %, 9.0 %, and 4.8 %, respectively. Different with the traditional coordinated operation, the proposed model sufficiently considers energy trading, payment bargaining, and information privacy. (2) The asymmetric Nash bargaining method can perform a more reasonable distribution of cooperative benefits by differentiating the contributions of the participants. The participants are motivated to strive to improve their bargaining power and gain more benefits. (3) The distributed solution algorithm features sound convergence for the operation cost minimization subproblem and the payment bargaining subproblem. Compared with the centralized solution mode, the computation time of the distributed method is reduced from 537 s to 376 s. Therefore, the distributed algorithm achieves highly efficient solutions while protecting the private information of each participant.

Consider that renewable energy sources (e.g., PV and WT) possess tremendous intermittency and uncertainty, and the fluctuation of renewable energy generates a substantial influence on the safe and economic operation of the energy system. Hence, to investigate energy-sharing transactions deeply, future work will sufficiently account for the volatility of renewable energy and explore the optimization operation of the integrated energy systems under uncertain conditions.

CRediT authorship contribution statement

Meng Yang: Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Yisheng Liu:** Supervision, Funding acquisition, Conceptualization. **Takuro Kobashi:** Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.scs.2024.105791.

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